# Exploratory study of Neural Operators as surrogate models for collective effects in the FCC-ee High Energy Booster

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M2 Internship

# Theoretical framework

# Theoretical Framework

*Neural Operators & Latent space decomposition*

**Neural Operators as surrogate models for collective effects**

$$\frac{\partial \rho}{\partial s} + \{\rho, H\} = \mathcal{C}[\rho]$$

Implies ($\mu$-parametrized) solution as

$$\rho_{s+\Delta s} = \mathcal{G}_\theta(\rho_s, \mu)\rho_s \approx e^{\Delta s \mathcal{L}_{\text{wake}}} e^{\Delta s \mathcal{L}_{\text{ext}}} \rho_s.$$

**Surrogate for nonlinear effects** In this context

$$e^{\Delta s \mathcal{L}_{\text{ext}}} \in GL(\mathbb{R}^N)$$

Is a linear operator, while

$$e^{\Delta s \mathcal{L}_{\text{wake}}} \in O(\mathbb{R}^N)$$

Is a nonlinear operator.

# Theoretical Framework

*Neural Operators & Latent space decomposition*

**Latent space decomposition:** Learning dynamics is performed in latent space of an Auto Encoder (AE)

$$z = \mathcal{E}_\phi(\mathcal{H}(\rho),\, \sigma, \langle x \rangle) \in \mathbb{R}^{256},\ \widehat{\mathcal{H}}(\rho) = \mathcal{D}_\psi(z).$$

The objective is $z_{t+1} = \mathcal{T}_\theta(z_t, \text{lattice}, \mu)$ and decode to phase-space projections.

**Neural Operators as surrogate models for collective effects**

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# Theoretical Framework

*Data dimensionality reduction with a CVAE*

## Conditional Variational Auto Encoder

The beam distribution is represented by 15 projected phase-space histograms

$$x = \mathcal{H}(\rho) \in \mathbb{R}^{15\times64\times64},$$

which are compressed into a probabilistic latent representation for some distribution $\mathcal{D}(\mu_\phi, \Sigma_\phi)$

$$q_\phi(z|x, a, d) = \mathcal{D}(\mu_\phi, \Sigma_\phi).$$

The conditioning variables (beam moments and machine-domain parameters) are $a = (\sigma, \langle x\rangle)$, $d = (\mu, \kappa)$, and provide sampling information for the latent space,

$$z = \mu + \exp\left(\frac{1}{2}\log\sigma^2\right) \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I),$$

## Encoder–Decoder

Encoder

$$(\mu, \log\sigma^2) = E_\phi(x, a, d)$$

Decoder

$$\hat{x} = D_\psi(z, d)$$

The decoder reconstructs all 15 normalized phase-space projections simultaneously,

$$\hat{x} = \widehat{\mathcal{H}}(\rho).$$

The latent variable $z \in \mathbb{R}^{256}$ provides a compact representation on which the neural operator performs time evolution.

# Theoretical Framework

## *Lattice Tokenization*

### Element embedding

Each accelerator element is described by

$$p_i = (L, \theta, k_1, k_2, \ldots) \in \mathbb{R}^7$$

together with its longitudinal coordinate $s_i$.The normalized element parameters are embedded through a multilayer perceptron

$$h_i^{(p)} = \mathrm{MLP}(p_i).$$

This produces a learnable semantic representation of every lattice component.

### Fourier positional encoding

Absolute position along the machine is encoded using multi-scale Fourier features

$$\gamma(s) = [\sin(\omega_k s), \cos(\omega_k s)]_{k=1}^{32},$$

where fourier components are logarithmically spaced. The final token is

$$h_i = \mathrm{MLP}(p_i) + W_p \gamma(s_i) \in \mathbb{R}^{512}.$$

This encoding captures both local optics and long-range machine periodicity.

# Neural Operators

# Neural Operators

*Graph Neural Operator formulation*

### From PDEs to operator learning

We consider a general elliptic PDE of the form

$$\mathcal{L}_a u(x) = f(x), \quad x \in D, \qquad u(x) = 0, \quad x \in \partial D.$$

For each coefficient field $a(x)$, the solution defines a mapping

$$\mathcal{F}^{\text{true}} : a(\cdot) \mapsto u(\cdot).$$

We aim to learn a discretization-invariant operator

$$\mathcal{F}_\theta \approx \mathcal{F}^{\text{true}} \quad \text{from samples } \{a_j, u_j\}.$$

### Green's function formalism

The solution operator can be written as an integral operator

$$u(x) = \int_D G_a(x, y)\, f(y)\, dy,$$

where $G_a$ is the Green's function depending on $a(x)$.
We replace it with a learnable kernel

$$u(x) = \int_D \kappa_\theta(x, y, a(x), a(y))\, u(y)\, dy.$$

This defines a data-driven Green's function approximation.

# Neural Operators

*Iterative kernel evolution & graph form*

## Iterative solver view

Instead of solving directly the differential equation, we iterate:

$$v_{t+1()}(x) = \sigma\left(W v_t(x) + \int_{B(x,r)} \kappa_\theta(x, y, a(x), a(y)) v_t(y)\, dy\right).$$

This defines a nonlinear implicit propagation scheme over function space.

## Graph discretization

On a discretization $P_K = \{x_i\}_{i=1}^K$:

$$v_{t+1()}(x) = \sigma\left(W v_t(x) + \frac{1}{|N(i)|} \sum_{j \in N(i)} \kappa_\theta(x_i, x_j, a_i, a_j)\, v_t(x_j)\right).$$

This is equivalent to message passing on a graph with learned edge kernels that can be directed enforcing **causality**:

$$m_{ij} = \kappa_\theta(e_{ij}) v_j, \quad e_{ij} = (x_i, x_j, a_i, a_j).$$

# Baseline Model

*Latent-space Transformer tracking*

## Transformer formulation in latent space

The baseline a standard autoregressive transformer operating on lattice tokens:

$$h_i = \mathrm{Tok}_\eta(p_i, s_i), \qquad h_i \in \mathbb{R}^{512}.$$

Latent evolution is modeled as:

$$z_{i+1} = \mathrm{Transformer}_\theta(z_i, h_{\le i}), \quad z_0 = \mathcal{E}_\phi(x, a, d).$$

This corresponds to:

$$z_N = z_0 + \sum_{i \le N} \Delta z_i,$$

with causal masking enforcing sequential dependence.

## Features

The attention mechanism is defined as:

$$\mathrm{A}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V.$$

it introduces key unphysical effects in the context of beam dynamics:

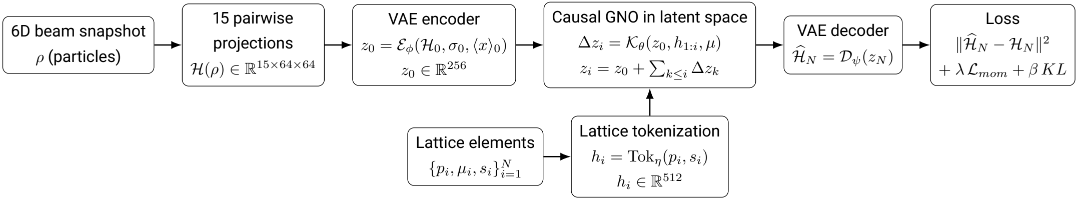
- global dense interactions (non-local mixing)
- weak inductive bias for spatial structure
- $O(N^2)$ complexity in lattice length

# Architecture

# Architecture

*CVAE + GNO*

## Proposed architecture




*Training data was generated using X-suite and PyHEADTAIL including CST data as the impedance model.*

# Architecture

*Graph Neural Operator configuration*

**Graph Neural Operator configuration**

| | |
|---|---|
| Hidden width | 512 |
| Operator layers | 4 |
| Token dimension | 512 |
| Element parameters | 7 |
| Fourier frequencies | 32 pairs (64 features) |
| Activation | GELU |
| Propagation | Kernel message passing |
| Tracking | Non-autoregressive |

# Architecture

## *Optimization objective*

### Total objective

The model is trained using a multi-objective loss

$$\boxed{\mathcal{L} = \mathcal{L}_{\mathrm{hist}} + \lambda_\sigma \mathcal{L}_\sigma + \lambda_c \mathcal{L}_{\mathrm{cent}} + \beta \mathcal{L}_{\mathrm{KL}}}$$

where

$$\mathcal{L}_{\mathrm{hist}} = \left\| \widehat{\mathcal{H}}_N - \mathcal{H}_N \right\|_2^2 ,$$

$$\mathcal{L}_\sigma = \left\| \widehat{\log \sigma} - \log \sigma \right\|_2^2 ,$$

$$\mathcal{L}_{\mathrm{cent}} = \|\hat{c} - c\|_2^2 .$$

The reconstruction loss is evaluated on normalized 15-channel pairwise beam projections.

### KL regularization

The latent posterior is constrained towards a unit Gaussian,

$$\mathcal{L}_{\mathrm{KL}} = \frac{1}{2} \sum_i \left( e^{\log \sigma_i^2} + \mu_i^2 - 1 - \log \sigma_i^2 \right) .$$

Provides the follwing properties:

- regularize the latent distribution;
- promote a smooth and continuous latent manifold;
- prevent overfitting and posterior collapse;
- improve interpolation and extrapolation;
- provide a stable latent space for the downstream GNO evolution.

# Preliminary Results

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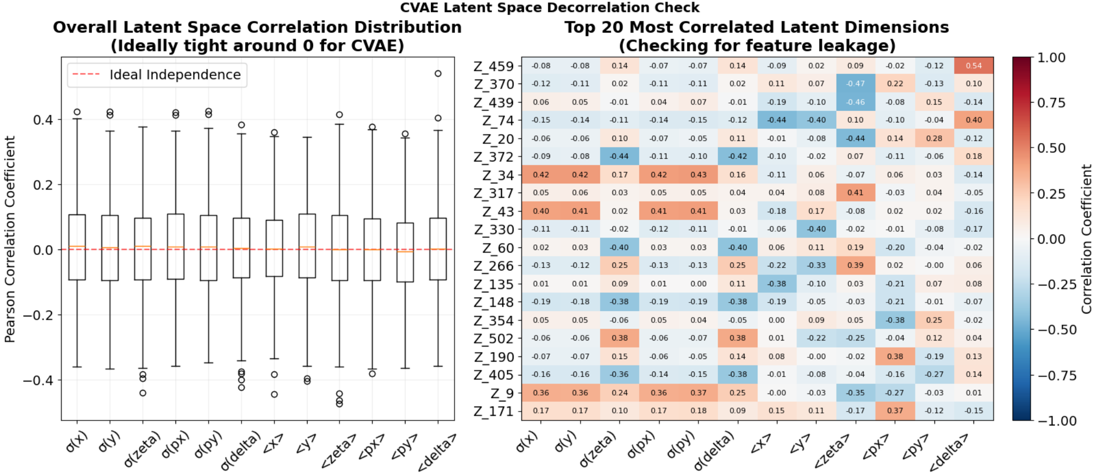
## *CVAE latent space reconstruction*



# Preliminary Results

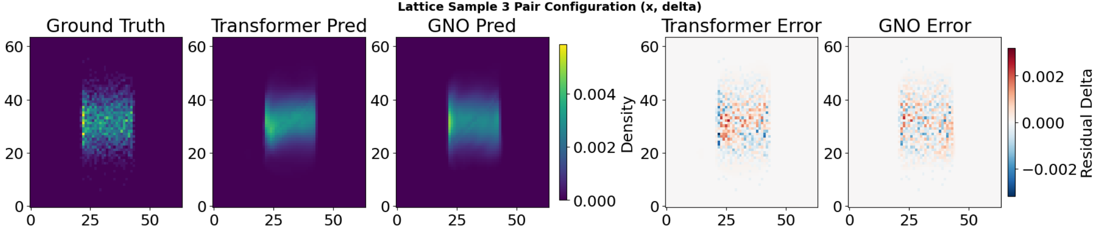
*CVAE latent space reconstruction*

**Key observations**

- Global latent–physics correlation is close to zero across almost all dimensions.
- The distribution is sharply centered around 0, indicating weak linear dependence between latent variables and physical summaries.
- Only a small subset of latent dimensions shows moderate correlation ($\sim 0.3$–$0.4$).
- These correlated modes are sparse and not structurally organized.
- No systematic leakage of physical variables into the latent representation is observed.

# Preliminary Results

*GNO vs Transformer latent space dynamics*



# Preliminary Results

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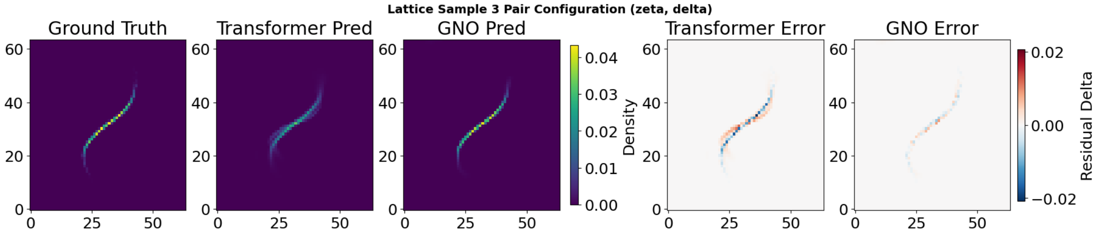

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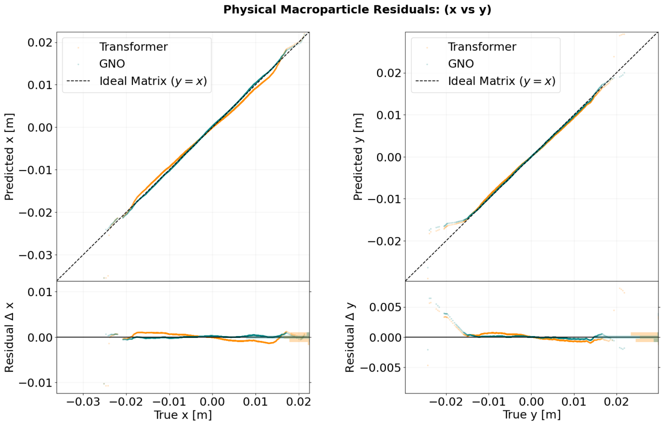
*GNO vs Transformer latent space dynamics*



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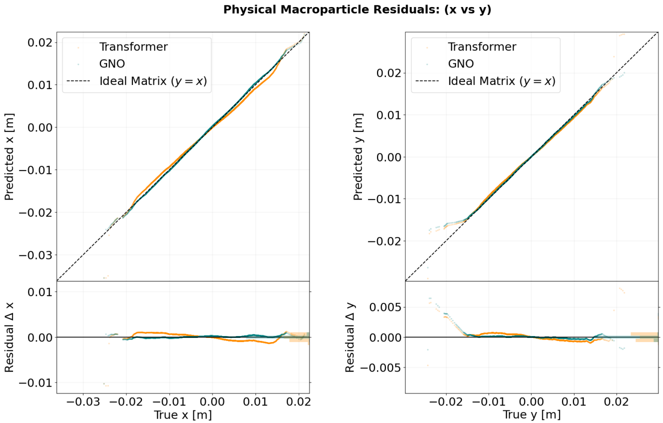
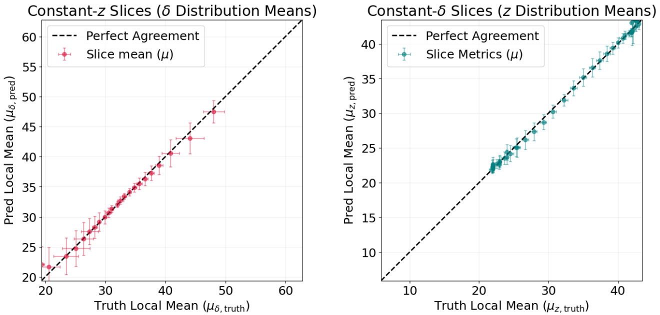
## *GNO vs Transformer latent space dynamics*



# Preliminary Results

*GNO vs Transformer latent space dynamics*

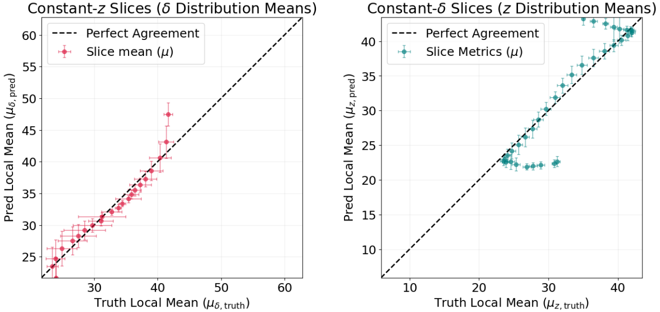
## GNO marginals exhibit causality



# Preliminary Results

*GNO vs Transformer latent space dynamics*

## Transformer marginals may produce artifacts due to non-causal attention




Gracias